

MDS220 Musculoskeletal System

Comprehensive Practice Examination • Lectures 1 to 22 • Single Best Answer

171 questions built from the course slides. All answer choices are written in full. Choose the single best answer.

SECTION A -- Histology, Cartilage, Bone, Integument, and Joint Introduction

1. Which anatomical plane divides the body into left and right portions?
 - (A) Transverse plane
 - (B) Sagittal plane
 - (C) Horizontal plane
 - (D) Coronal plane
 - (E) Oblique plane
2. The default view of a computed tomography scan corresponds to which plane?
 - (A) Transverse plane
 - (B) Sagittal plane
 - (C) Oblique plane
 - (D) Median plane
 - (E) Coronal plane
3. A movement that returns the thumb from opposition to its anatomical position is called?
 - (A) Retraction
 - (B) Opposition
 - (C) Protraction
 - (D) Reposition
 - (E) Circumduction
4. A peg-in-socket fibrous joint between a tooth root and its alveolus is a?
 - (A) Synchondrosis
 - (B) Symphysis
 - (C) Gomphosis
 - (D) Suture
 - (E) Syndesmosis
5. The interosseous membrane between the tibia and fibula is which kind of fibrous joint?
 - (A) Symphysis
 - (B) Synchondrosis
 - (C) Gomphosis
 - (D) Suture
 - (E) Syndesmosis
6. The epiphyseal plate joining the diaphysis and epiphysis is which type of joint?
 - (A) Gomphosis
 - (B) Synchondrosis
 - (C) Syndesmosis
 - (D) Symphysis
 - (E) Suture
7. The pubic symphysis and intervertebral discs are examples of which joint type?
 - (A) Gomphosis
 - (B) Synchondrosis
 - (C) Syndesmosis
 - (D) Suture
 - (E) Symphysis
8. Which synovial joint type permits only flexion and extension around a single axis?
 - (A) Hinge joint

- (B) Saddle joint
- (C) Plane joint
- (D) Condylloid joint
- (E) Pivot joint

9. The carpometacarpal joint of the thumb is classified as which synovial joint?

- (A) Pivot joint
- (B) Plane joint
- (C) Saddle joint
- (D) Condylloid joint
- (E) Hinge joint

10. A meniscus, present in the knee, is composed of which tissue?

- (A) Dense tendon
- (B) Elastic cartilage
- (C) Loose connective
- (D) Fibrocartilage
- (E) Hyaline cartilage

11. Stretching or tearing of a muscle or tendon is defined as a?

- (A) Strain injury
- (B) Bursitis
- (C) Dislocation
- (D) Subluxation
- (E) Sprain injury

12. Inflammation of a fluid-filled sac that reduces friction around a joint is called?

- (A) Strain injury
- (B) Bursitis
- (C) Tendinitis
- (D) Dislocation
- (E) Sprain injury

13. Which resident connective-tissue cell has a large oval euchromatic nucleus and builds the matrix?

- (A) Adipocyte
- (B) Mast cell
- (C) Plasma cell
- (D) Fibroblast
- (E) Macrophage

14. A cell with a clock-face nucleus and prominent Golgi zone that secretes antibodies is the?

- (A) Neutrophil
- (B) Mast cell
- (C) Plasma cell
- (D) Fibroblast
- (E) Macrophage

15. Which cell stores histamine and heparin in metachromatic basophilic granules?

- (A) Fibroblast
- (B) Plasma cell
- (C) Eosinophil
- (D) Adipocyte
- (E) Mast cell

16. The macrophage found in the epidermis of the skin is specifically named?

- (A) Dust cell
- (B) Osteoclast cell
- (C) Langerhans cell
- (D) Microglial cell

(E) Kupffer cell

17. The first phagocyte to leave the bloodstream and form pus at infection sites is the?

- (A) Fibroblast
- (B) Neutrophil
- (C) Lymphocyte
- (D) Plasma cell
- (E) Eosinophil

18. Which leukocyte increases after parasitic infection and binds antigen-antibody complexes on parasites?

- (A) Eosinophil
- (B) Plasma cell
- (C) Neutrophil
- (D) Mast cell
- (E) Lymphocyte

19. The ground substance of connective tissue is composed mainly of proteoglycans and?

- (A) Elastic fibres
- (B) Calcium salts
- (C) Keratin protein
- (D) Glycoproteins
- (E) Collagen fibres

20. Type IV collagen is characteristically found in which location?

- (A) Large artery wall
- (B) Hyaline cartilage
- (C) Tendon tissue
- (D) Bone matrix
- (E) Basement membrane

21. Reticular fibres are made of which collagen and are demonstrated by which stain?

- (A) Type two orcein
- (B) Type one eosin
- (C) Type four gold
- (D) Type three silver
- (E) Type five iron

22. Marfan syndrome arises from a genetic deficiency of which protein?

- (A) Dystrophin protein
- (B) Type one collagen
- (C) Fibrillin one
- (D) Tyrosinase enzyme
- (E) Keratin filament

23. Vascular Ehlers-Danlos syndrome, with vessel and organ rupture, involves which collagen?

- (A) Type five collagen
- (B) Type one collagen
- (C) Type two collagen
- (D) Type four collagen
- (E) Type three collagen

24. Wharton's jelly of the umbilical cord is an example of which connective tissue?

- (A) Mucous connective
- (B) Elastic connective
- (C) Reticular connective
- (D) Dense regular tissue
- (E) Adipose connective

25. The multiadhesive glycoproteins that bind cells to matrix through integrins are chiefly?

- (A) Tropocollagen unit
- (B) Fibronectin
- (C) Hyaluronan chain
- (D) Chondroitin sulfate
- (E) Aggrecan core

26. Brown adipose tissue generates heat because it is especially rich in?

- (A) Lysosomes
- (B) Ribosomes
- (C) Golgi bodies
- (D) Peroxisomes
- (E) Mitochondria

27. In skeletal muscle a triad consists of one transverse tubule flanked by?

- (A) Two dense bodies
- (B) Two sarcosomes
- (C) Two cisternae
- (D) One caveola
- (E) One cisterna

28. Which band of the sarcomere contains only thin actin filaments?

- (A) The H zone region
- (B) The anisotropic band
- (C) The M line zone
- (D) The isotropic band
- (E) The Z disc zone

29. Curare causes flaccid paralysis by binding to which site?

- (A) Postsynaptic receptor
- (B) Calcium pump site
- (C) Presynaptic terminal
- (D) Actin binding site
- (E) Transverse tubule

30. Botulinum toxin produces flaccid paralysis by acting at which site?

- (A) Transverse tubule
- (B) Presynaptic membrane
- (C) Calcium pump site
- (D) Postsynaptic receptor
- (E) Troponin binding

31. Myasthenia gravis is an autoimmune disease with antibodies against which target?

- (A) Sodium channel gate
- (B) Acetylcholine receptor
- (C) Presynaptic terminal
- (D) Actin filament core
- (E) Calcium storage pump

32. The Golgi tendon organ primarily monitors which variable?

- (A) Joint position
- (B) Blood oxygen
- (C) Skin stretching
- (D) Muscle length
- (E) Muscle tension

33. The knee-jerk reflex depends on which sensory structure detecting stretch?

- (A) Merkel receptor
- (B) Pacinian corpuscle

- (C) Ruffini ending
- (D) Muscle spindle
- (E) Golgi tendon organ

34. Gap junctions inside intercalated discs of cardiac muscle allow?

- (A) Fat breakdown
- (B) Toxin removal
- (C) Ionic coupling
- (D) Oxygen storage
- (E) Calcium capture

35. The calcium-storing junction of cardiac muscle at the Z-line is called a?

- (A) Diad structure
- (B) Sarcomere unit
- (C) Caveola pocket
- (D) Triad structure
- (E) Dense body plate

36. In smooth muscle, thin filaments attach not to a Z-disc but to?

- (A) Triad junctions
- (B) Intercalated discs
- (C) Dense bodies
- (D) Terminal cisternae
- (E) Motor end plates

37. In smooth muscle, sarcolemmal invaginations that trap calcium are called?

- (A) Diads
- (B) Caveolae
- (C) Sarcosomes
- (D) Triads
- (E) Cisternae

38. Which muscle type increases its work output by hyperplasia (cell division)?

- (A) Cardiac muscle
- (B) Skeletal muscle
- (C) Voluntary muscle
- (D) Smooth muscle
- (E) Striated muscle

39. Which connective-tissue layer directly surrounds a single skeletal muscle fibre?

- (A) Endomysium
- (B) Epimysium
- (C) Deep fascia
- (D) Perimysium
- (E) Periosteum

40. The regenerating cell lying beneath the basal lamina of a skeletal muscle fibre is the?

- (A) Pericyte cell
- (B) Osteocyte cell
- (C) Fibrocyte cell
- (D) Myofibroblast
- (E) Satellite cell

41. Which cartilage type has type one collagen and lacks a perichondrium?

- (A) Articular cartilage
- (B) Costal cartilage
- (C) Fibrocartilage
- (D) Elastic cartilage
- (E) Hyaline cartilage

42. Elastic cartilage is identified histologically by which special stain?
- (A) Periodic-acid stain
 - (B) Orcein stain
 - (C) Silver stain
 - (D) Hematoxylin only
 - (E) Masson stain
43. Cartilage grows in length by which mechanism?
- (A) Interstitial growth
 - (B) Appositional growth
 - (C) Perichondral growth
 - (D) Endochondral growth
 - (E) Intramembranous way
44. New cartilage added to the surface by perichondrial chondroblasts is which growth?
- (A) Intramembranous way
 - (B) Interstitial growth
 - (C) Epiphyseal growth
 - (D) Endochondral growth
 - (E) Appositional growth
45. Which hormone induces cartilage growth and matrix formation?
- (A) Cortisone
 - (B) Hydrocortisone
 - (C) Aldosterone
 - (D) Somatotropin
 - (E) Estradiol
46. The territorial matrix immediately around a chondrocyte stains basophilic because it is rich in?
- (A) Elastic fibre
 - (B) Chondroitin sulfate
 - (C) Keratin protein
 - (D) Calcium phosphate
 - (E) Type one collagen
47. Deficiency of vitamin C deforms the epiphyseal plate because it is required for?
- (A) Fat storage
 - (B) Calcium uptake
 - (C) Matrix synthesis
 - (D) Melanin making
 - (E) Elastin recoil
48. The organic matrix of bone is composed chiefly of which collagen?
- (A) Type three collagen
 - (B) Type four collagen
 - (C) Type two collagen
 - (D) Type one collagen
 - (E) Type five collagen
49. The inorganic component of bone matrix is mainly?
- (A) Glycoprotein
 - (B) Tropocollagen
 - (C) Keratan sulfate
 - (D) Chondroitin salt
 - (E) Hydroxyapatite
50. Osteoclasts are derived from which precursor lineage?
- (A) Monocyte lineage
 - (B) Chondroblast line

- (C) Osteoblast lineage
- (D) Endothelial line
- (E) Fibroblast lineage

51. Parathyroid hormone affects bone by producing which effect?

- (A) Inhibit osteoclasts
- (B) Remove marrow
- (C) Stimulate osteoclasts
- (D) Build cartilage
- (E) Halt remodeling

52. The structural unit of compact bone centred on a central canal is the?

- (A) Cement line
- (B) Trabecula plate
- (C) Lacuna space
- (D) Osteon system
- (E) Canaliculus tube

53. Transverse canals connecting central canals and carrying vessels are named?

- (A) Volkmann canals
- (B) Lamellae
- (C) Canaliculi
- (D) Haversian canals
- (E) Howship lacunae

54. Immature woven bone differs from mature lamellar bone by having?

- (A) Regular fibres
- (B) No osteocytes
- (C) Greater stiffness
- (D) No osteoblasts
- (E) Irregular fibres

55. Interstitial lamellae in compact bone represent?

- (A) Marrow spaces
- (B) Osteon remnants
- (C) New periosteum
- (D) Volkmann lines
- (E) Growth plates

56. In endochondral ossification the template for the future bone is?

- (A) Loose mesenchyme
- (B) Elastic cartilage
- (C) Hyaline cartilage
- (D) Woven bone
- (E) Fibrocartilage

57. During bone repair, osteoblasts producing osteoid corresponds to which phase?

- (A) Resorption phase
- (B) Remodeling phase
- (C) Fibrocartilage phase
- (D) Hematoma phase
- (E) Bony callus phase

58. According to the slides, cancellous bone unites after fracture in approximately?

- (A) Four to six weeks
- (B) Two to three days
- (C) One to two weeks
- (D) Six to nine months
- (E) Eight to twelve weeks

59. In osteoporosis, the bone imbalance is best described as?
- (A) Excess collagen forms
 - (B) No mineralisation occurs
 - (C) No resorption occurs
 - (D) Resorption exceeds formation
 - (E) Formation exceeds resorption
60. The epidermis develops embryologically from?
- (A) Lateral mesoderm
 - (B) Surface ectoderm
 - (C) Neural tube
 - (D) Notochord
 - (E) Paraxial mesoderm
61. Which epidermal cell is a light-touch receptor located in the stratum basale?
- (A) Merkel cell
 - (B) Keratinocyte
 - (C) Langerhans cell
 - (D) Melanocyte cell
 - (E) Fibroblast cell
62. Langerhans cells of the epidermis function as?
- (A) Pigment producers
 - (B) Antigen presenters
 - (C) Water barriers
 - (D) Keratin makers
 - (E) Touch receptors
63. Which epidermal layer is present in thick skin but absent in thin skin?
- (A) Stratum basale
 - (B) Stratum granulosum
 - (C) Stratum lucidum
 - (D) Stratum spinosum
 - (E) Stratum corneum
64. Keratohyalin granules that mark the transition toward dead cells appear in which layer?
- (A) Stratum basale
 - (B) Stratum lucidum
 - (C) Stratum corneum
 - (D) Stratum spinosum
 - (E) Stratum granulosum
65. Melanocytes of the epidermis are derived from which embryonic cells?
- (A) Lateral mesoderm cells
 - (B) Paraxial mesoderm cells
 - (C) Surface ectoderm cells
 - (D) Neural crest cells
 - (E) Endoderm cells
66. The papillary layer of the dermis is composed of which tissue?
- (A) Dense regular tissue
 - (B) Loose connective tissue
 - (C) Dense irregular tissue
 - (D) Reticular tissue
 - (E) Adipose tissue
67. The reticular layer of the dermis is composed of which tissue?
- (A) Elastic tissue
 - (B) Adipose tissue

- (C) Loose connective tissue
- (D) Dense regular tissue
- (E) Dense irregular tissue

68. Which sweat gland opens into a hair follicle and is found in the axilla?

- (A) Eccrine gland
- (B) Mammary gland
- (C) Sebaceous gland
- (D) Apocrine gland
- (E) Ceruminous gland

69. Eccrine sweat glands function primarily as organs of?

- (A) Thermoregulation
- (B) Sensory reception
- (C) Pigment production
- (D) Sebum secretion
- (E) Antigen presenting

70. Sebaceous glands secrete sebum into which structure?

- (A) Nail matrix
- (B) Blood vessel
- (C) Hair follicle
- (D) Dermal papilla
- (E) Sweat gland duct

71. The arrector pili muscle is which muscle type, with what nerve supply?

- (A) Skeletal autonomic muscle
- (B) Smooth sympathetic muscle
- (C) Cardiac autonomic muscle
- (D) Smooth parasympathetic muscle
- (E) Skeletal somatic muscle

72. The natural pigment and moisture of a hair reside in which zone of the shaft?

- (A) Hair cuticle
- (B) Hair medulla
- (C) Hair root
- (D) Hair cortex
- (E) Hair papilla

SECTION B -- Gross Anatomy (Back, Upper and Lower Limb, Joints, Vertebral Column)

73. Dorsal scapular nerve injury causes weakness of which muscle?

- (A) Splenius capitis
- (B) Latissimus dorsi
- (C) Trapezius muscle
- (D) Serratus anterior
- (E) Rhomboid major

74. The spinal accessory nerve supplies which back muscle?

- (A) Trapezius muscle
- (B) Rhomboid major
- (C) Latissimus dorsi
- (D) Levator scapulae
- (E) Serratus posterior

75. The triangle of auscultation is bounded by trapezius, rhomboid major and which muscle?

- (A) Teres major muscle

- (B) Splenius capitis
- (C) Latissimus dorsi
- (D) Levator scapulae
- (E) Serratus posterior

- 76.** The erector spinae and other deep back muscles are innervated by which nerve rami?
- (A) Sympathetic trunk
 - (B) Ventral primary rami
 - (C) Phrenic branches
 - (D) Dorsal primary rami
 - (E) Gray communicantes
- 77.** The suboccipital triangle is bounded by rectus capitis posterior major and which two muscles?
- (A) Only the inferior oblique
 - (B) Superior and inferior obliques
 - (C) Inferior oblique and semispinalis
 - (D) Only the superior oblique
 - (E) Superior oblique and splenius
- 78.** The contents of the suboccipital triangle include the vertebral artery and which nerve?
- (A) Accessory nerve
 - (B) Lesser occipital nerve
 - (C) Suboccipital nerve
 - (D) Great auricular nerve
 - (E) Greater occipital nerve
- 79.** The greater occipital nerve arises from the dorsal ramus of which spinal nerve?
- (A) Second cervical nerve
 - (B) Fourth cervical nerve
 - (C) First cervical nerve
 - (D) Third cervical nerve
 - (E) Fifth cervical nerve
- 80.** Which four muscles make up the rotator cuff?
- (A) Pectoralis latissimus and teres
 - (B) Trapezius and the two rhomboids
 - (C) Biceps triceps and coracobrachialis
 - (D) Deltoid teres major and biceps
 - (E) Supraspinatus infraspinatus teres minor subscapularis
- 81.** The axillary nerve and posterior circumflex humeral artery pass through which space?
- (A) Triangular space
 - (B) Carpal tunnel
 - (C) Cubital fossa
 - (D) Quadrangular space
 - (E) Deltopectoral triangle
- 82.** The deltoid, chief abductor beyond fifteen degrees, is supplied by which nerve?
- (A) Suprascapular nerve
 - (B) Musculocutaneous nerve
 - (C) Axillary nerve
 - (D) Long thoracic nerve
 - (E) Radial nerve
- 83.** Supraspinatus, which initiates abduction, is supplied by which nerve?
- (A) Dorsal scapular nerve
 - (B) Axillary nerve
 - (C) Upper subscapular nerve
 - (D) Long thoracic nerve

(E) Suprascapular nerve

84. The scapulohumeral rhythm during arm abduction has a ratio of about?

- (A) One to three
- (B) One to one
- (C) Five to one
- (D) Three to one
- (E) Two to three

85. Which muscle forms the anterior axillary fold?

- (A) Teres major muscle
- (B) Pectoralis major
- (C) Serratus anterior
- (D) Latissimus dorsi
- (E) Deltoid muscle

86. The deltopectoral triangle is bounded by the clavicle, deltoid and which muscle border?

- (A) Pectoralis major
- (B) Pectoralis minor
- (C) Latissimus dorsi
- (D) Serratus anterior
- (E) Subclavius muscle

87. The cephalic vein pierces the clavipectoral fascia to drain into which vein?

- (A) Brachial vein
- (B) Axillary vein
- (C) Basilic vein
- (D) Internal jugular
- (E) Subclavian vein

88. After axillary lymph-node clearance the lower pectoralis major is paralysed. Which nerve was injured?

- (A) Thoracodorsal nerve
- (B) Musculocutaneous nerve
- (C) Lower subscapular nerve
- (D) Lateral pectoral nerve
- (E) Medial pectoral nerve

89. Serratus anterior, which protracts the scapula, is supplied by which nerve?

- (A) Thoracodorsal nerve
- (B) Dorsal scapular nerve
- (C) Long thoracic nerve
- (D) Medial pectoral nerve
- (E) Lateral pectoral nerve

90. The anterior (flexor) compartment of the arm is supplied by which nerve?

- (A) Ulnar nerve
- (B) Radial nerve
- (C) Axillary nerve
- (D) Musculocutaneous nerve
- (E) Median nerve

91. The radial nerve descends the arm within which groove of the humerus?

- (A) Bicipital groove
- (B) Ulnar groove
- (C) Subscapular groove
- (D) Deltoid line
- (E) Radial groove

- 92.** The main artery passing through the cubital fossa deep to the bicipital aponeurosis is the?
- (A) Brachial artery
 - (B) Radial nerve
 - (C) Cephalic vein
 - (D) Ulnar nerve
 - (E) Median cubital vein
- 93.** The lateral boundary of the cubital fossa is formed by which muscle?
- (A) Pronator teres muscle
 - (B) Brachioradialis muscle
 - (C) Biceps brachii tendon
 - (D) Supinator muscle
 - (E) Brachialis muscle
- 94.** The biceps brachii tendon inserts onto which bony landmark?
- (A) Ulnar tuberosity
 - (B) Coronoid process
 - (C) Radial styloid
 - (D) Radial tuberosity
 - (E) Olecranon process
- 95.** The four superficial anterior forearm flexors share a common origin at the?
- (A) Radial head
 - (B) Coronoid process
 - (C) Medial epicondyle
 - (D) Lateral epicondyle
 - (E) Olecranon process
- 96.** The posterior forearm extensors arise mainly from which epicondyle?
- (A) Medial epicondyle
 - (B) Radial styloid
 - (C) Lateral epicondyle
 - (D) Coronoid process
 - (E) Olecranon process
- 97.** Pronator teres pronates and flexes the forearm and is supplied by which nerve?
- (A) Deep radial nerve
 - (B) Anterior interosseous nerve
 - (C) Ulnar nerve
 - (D) Radial nerve
 - (E) Median nerve
- 98.** Saturday night palsy from radial nerve compression classically produces?
- (A) Ape hand
 - (B) Wrist drop
 - (C) Claw hand
 - (D) Foot drop
 - (E) Winged scapula
- 99.** A Colles fracture involves the distal end of which bone?
- (A) Radius bone
 - (B) Humerus bone
 - (C) Ulna bone
 - (D) Scaphoid bone
 - (E) Lunate bone
- 100.** The floor of the anatomical snuffbox is formed by the scaphoid and which bone?
- (A) Capitate bone
 - (B) Hamate bone

- (C) Lunate bone
- (D) Trapezium bone
- (E) Trapezoid bone

101. The content that passes through the anatomical snuffbox is the?

- (A) Second part of radial artery
- (B) Cephalic vein origin
- (C) Basilic vein ending
- (D) First part of ulnar artery
- (E) Median nerve branch

102. The central compartment of the palm contains the flexor tendons, lumbricals and which arch?

- (A) Radial artery loop
- (B) Ulnar nerve deep branch
- (C) Superficial palmar arch
- (D) Deep palmar arch
- (E) Thenar muscle group

103. The palmar aponeurosis is continuous proximally with which tendon?

- (A) Flexor digitorum superficialis tendon
- (B) Flexor pollicis longus tendon
- (C) Flexor carpi radialis tendon
- (D) Pronator quadratus tendon
- (E) Palmaris longus tendon

104. The extensor retinaculum forms how many osseofibrous tunnels for the extensor tendons?

- (A) Nine tunnels
- (B) Six tunnels
- (C) Four tunnels
- (D) Three tunnels
- (E) Two tunnels

105. The sternoclavicular joint is classified as which synovial type?

- (A) Pivot joint
- (B) Ball and socket
- (C) Plane joint
- (D) Saddle joint
- (E) Hinge joint

106. The metacarpophalangeal joints are classified as which synovial type?

- (A) Pivot joint
- (B) Hinge joint
- (C) Saddle joint
- (D) Condylloid joint
- (E) Plane joint

107. The acromioclavicular joint is classified as which synovial type?

- (A) Pivot joint
- (B) Plane joint
- (C) Hinge joint
- (D) Ball and socket
- (E) Saddle joint

108. The glenohumeral joint is deepened by which fibrocartilaginous rim?

- (A) Glenoid labrum
- (B) Coronoid rim
- (C) Meniscus rim
- (D) Annular ligament
- (E) Articular disc

109. The ligament injured at the proximal radioulnar joint, encircling the radial head, is the?

- (A) Radial collateral ligament
- (B) Interosseous membrane
- (C) Annular ligament
- (D) Ulnar collateral ligament
- (E) Palmar radioulnar ligament

110. Which muscle is the chief extensor and lateral rotator of the thigh at the hip?

- (A) Gluteus medius
- (B) Iliopsoas muscle
- (C) Piriformis muscle
- (D) Gluteus minimus
- (E) Gluteus maximus

111. Gluteus medius and minimus abduct and medially rotate the thigh; they are supplied by which nerve?

- (A) Superior gluteal nerve
- (B) Sciatic nerve
- (C) Obturator nerve
- (D) Inferior gluteal nerve
- (E) Femoral nerve

112. Injury to which nerve produces a positive Trendelenburg sign?

- (A) Femoral nerve
- (B) Inferior gluteal nerve
- (C) Sciatic nerve
- (D) Superior gluteal nerve
- (E) Obturator nerve

113. The hamstring muscles arise from which bony landmark?

- (A) Pubic tubercle
- (B) Lesser trochanter
- (C) Greater trochanter
- (D) Iliac crest
- (E) Ischial tuberosity

114. The hamstrings are innervated mainly by which nerve?

- (A) Common fibular nerve
- (B) Tibial nerve
- (C) Femoral nerve
- (D) Inferior gluteal nerve
- (E) Obturator nerve

115. The recommended safe line for gluteal intramuscular injection runs from the posterior superior iliac spine to the?

- (A) Iliac crest
- (B) Ischial tuberosity
- (C) Greater trochanter
- (D) Lesser trochanter
- (E) Pubic tubercle

116. The most superficial major nerve in the popliteal fossa is the?

- (A) Saphenous nerve
- (B) Sural nerve
- (C) Common fibular nerve
- (D) Tibial nerve
- (E) Obturator nerve

117. The popliteal fossa is bounded superolaterally by which muscle?

- (A) Biceps femoris
- (B) Semimembranosus
- (C) Lateral gastrocnemius
- (D) Semitendinosus
- (E) Medial gastrocnemius

118. The small saphenous vein terminates by draining into which vein?

- (A) Peroneal vein
- (B) Popliteal vein
- (C) Anterior tibial vein
- (D) Femoral vein
- (E) Great saphenous vein

119. The anterior compartment of the thigh is supplied by which nerve?

- (A) Obturator nerve
- (B) Sciatic nerve
- (C) Tibial nerve
- (D) Superior gluteal nerve
- (E) Femoral nerve

120. The medial (adductor) compartment of the thigh is chiefly supplied by which nerve?

- (A) Superior gluteal nerve
- (B) Sciatic nerve
- (C) Obturator nerve
- (D) Femoral nerve
- (E) Tibial nerve

121. The femoral triangle is bounded superiorly by the inguinal ligament, laterally by sartorius and medially by?

- (A) Adductor magnus
- (B) Vastus medialis
- (C) Adductor longus
- (D) Pectineus muscle
- (E) Gracilis muscle

122. The adductor (subsartorial) canal contains the femoral vessels, the nerve to vastus medialis and the?

- (A) Sciatic nerve
- (B) Saphenous nerve
- (C) Obturator nerve
- (D) Tibial nerve
- (E) Sural nerve

123. Which muscle both flexes the hip and extends the knee?

- (A) Vastus lateralis
- (B) Sartorius muscle
- (C) Vastus medialis
- (D) Rectus femoris
- (E) Iliacus muscle

124. The femoral canal, the medial compartment of the femoral sheath, normally contains?

- (A) The great saphenous vein
- (B) The obturator nerve
- (C) The femoral artery
- (D) The femoral nerve
- (E) Loose tissue and lymphatics

125. The anterior leg compartment dorsiflexors are supplied by which nerve?

- (A) Deep fibular nerve

- (B) Sural nerve
- (C) Saphenous nerve
- (D) Superficial fibular nerve
- (E) Tibial nerve

126. The lateral leg compartment everters are supplied by which nerve?

- (A) Superficial fibular nerve
- (B) Sural nerve
- (C) Deep fibular nerve
- (D) Tibial nerve
- (E) Saphenous nerve

127. The posterior leg compartment plantarflexors are supplied by which nerve?

- (A) Deep fibular nerve
- (B) Saphenous nerve
- (C) Superficial fibular nerve
- (D) Sural nerve
- (E) Tibial nerve

128. Which muscle produces both inversion and plantarflexion of the foot?

- (A) Fibularis longus
- (B) Tibialis anterior
- (C) Fibularis brevis
- (D) Tibialis posterior
- (E) Extensor hallucis longus

129. The great saphenous vein terminates by draining into which vein?

- (A) Anterior tibial vein
- (B) Femoral vein
- (C) Posterior tibial vein
- (D) Popliteal vein
- (E) Peroneal vein

130. The medial longitudinal arch of the foot is chiefly supported by which ligament?

- (A) Deltoid ligament
- (B) Long plantar ligament
- (C) Spring ligament
- (D) Bifurcate ligament
- (E) Short plantar ligament

131. The transverse tarsal joint permits which pair of movements?

- (A) Rotation and gliding
- (B) Abduction and adduction
- (C) Inversion and eversion
- (D) Flexion and extension
- (E) Pronation and supination

132. The hip joint is classified as which synovial type?

- (A) Plane joint
- (B) Hinge joint
- (C) Saddle joint
- (D) Ball and socket joint
- (E) Pivot joint

133. The strongest hip ligament, resisting hyperextension, is the?

- (A) Iliofemoral ligament
- (B) Pubofemoral ligament
- (C) Ischiofemoral ligament
- (D) Transverse ligament

(E) Ligament of head

134. The ligament that resists posterior hip dislocation, torn in dashboard injuries, is the?

- (A) Pubofemoral ligament
- (B) Ischiofemoral ligament
- (C) Ligament of head
- (D) Transverse ligament
- (E) Iliofemoral ligament

135. The anterior cruciate ligament prevents which abnormal tibial movement?

- (A) Posterior sliding
- (B) Lateral sliding
- (C) Rotational locking
- (D) Medial sliding
- (E) Anterior sliding

136. The posterior cruciate ligament prevents which abnormal tibial movement?

- (A) Medial sliding
- (B) Anterior sliding
- (C) Rotational locking
- (D) Lateral sliding
- (E) Posterior sliding

137. The knee menisci are composed of which tissue?

- (A) Dense tendon
- (B) Elastic cartilage
- (C) Loose connective tissue
- (D) Fibrocartilage
- (E) Hyaline cartilage

138. The medial meniscus is firmly attached to which ligament, so both are injured together?

- (A) Posterior cruciate ligament
- (B) Anterior cruciate ligament
- (C) Tibial collateral ligament
- (D) Patellar ligament
- (E) Fibular collateral ligament

139. Unlocking of the fully extended knee is produced by which muscle?

- (A) Popliteus muscle
- (B) Semitendinosus muscle
- (C) Sartorius muscle
- (D) Quadriceps femoris
- (E) Gastrocnemius muscle

140. The ankle (talocrural) joint is classified as which synovial type?

- (A) Saddle joint
- (B) Hinge joint
- (C) Ball and socket joint
- (D) Pivot joint
- (E) Plane joint

141. A neck-of-talus fracture most likely injures which ligament?

- (A) Deltoid ligament
- (B) Posterior tibiofibular ligament
- (C) Anterior talofibular ligament
- (D) Calcaneofibular ligament
- (E) Spring ligament

142. Infection in which bursa cannot spread to the knee joint cavity?

- (A) Suprapatellar bursa
- (B) Gastrocnemius bursa
- (C) Deep infrapatellar bursa
- (D) Semimembranosus bursa
- (E) Popliteus bursa

143. The anterior longitudinal ligament chiefly prevents which movement of the vertebral column?

- (A) Hyperflexion
- (B) Hyperextension
- (C) Rotation
- (D) Lateral bending
- (E) Gliding

144. Which ligament helps prevent posterior herniation of the nucleus pulposus?

- (A) Supraspinous ligament
- (B) Ligamentum flavum
- (C) Anterior longitudinal ligament
- (D) Nuchal ligament
- (E) Posterior longitudinal ligament

145. The elastic ligament that limits abrupt flexion between adjacent laminae is the?

- (A) Ligamentum flavum
- (B) Posterior longitudinal ligament
- (C) Interspinous ligament
- (D) Anterior longitudinal ligament
- (E) Supraspinous ligament

146. The zygapophyseal (facet) joints are classified as which synovial type?

- (A) Saddle joint
- (B) Hinge joint
- (C) Plane joint
- (D) Pivot joint
- (E) Ball and socket joint

147. The median atlantoaxial joint, allowing head rotation, is which synovial type?

- (A) Pivot joint
- (B) Ball and socket joint
- (C) Plane joint
- (D) Saddle joint
- (E) Hinge joint

148. The dens of the axis is held in place by which ligament?

- (A) Nuchal ligament
- (B) Anterior longitudinal ligament
- (C) Supraspinous ligament
- (D) Ligamentum flavum
- (E) Transverse ligament of atlas

149. Herniation of the nucleus pulposus most commonly occurs in which direction?

- (A) Anteriorly
- (B) Superiorly
- (C) Laterally only
- (D) Posterolaterally
- (E) Directly posterior

150. Sciatica from a lumbar disc herniation most often compresses which nerve components?

- (A) The second and third lumbar
- (B) The fifth lumbar or first sacral
- (C) The first thoracic

- (D) The fourth sacral only
- (E) The first lumbar only

151. An abnormal lateral curvature of the vertebral column is termed?

- (A) Stenosis
- (B) Scoliosis
- (C) Kyphosis
- (D) Spondylosis
- (E) Lordosis

152. The primary curvatures of the vertebral column, present in the fetus, are the?

- (A) Lumbar and sacral curves
- (B) Cervical and lumbar lordoses
- (C) Only the thoracic curve
- (D) Thoracic and sacral kyphoses
- (E) Cervical and thoracic curves

SECTION C -- Embryology and Development

153. The nucleus pulposus of the intervertebral disc is a remnant of the?

- (A) Neural crest
- (B) Dermatome
- (C) Myotome
- (D) Sclerotome
- (E) Notochord

154. Vertebrae and ribs are derived from which part of the somite?

- (A) Dermatome
- (B) Notochord
- (C) Sclerotome
- (D) Myotome
- (E) Neural crest

155. Which bone belongs to the cartilaginous neurocranium (chondrocranium)?

- (A) Sphenoid bone
- (B) Frontal bone
- (C) Nasal bone
- (D) Parietal bone
- (E) Zygomatic bone

156. Which bone belongs to the membranous neurocranium?

- (A) Sphenoid bone
- (B) Petrous temporal
- (C) Frontal bone
- (D) Occipital base
- (E) Ethmoid bone

157. The malleus and incus develop from the cartilage of which pharyngeal arch?

- (A) Fourth pharyngeal arch
- (B) Sixth pharyngeal arch
- (C) Second pharyngeal arch
- (D) Third pharyngeal arch
- (E) First pharyngeal arch

158. The stapes and styloid process develop from the cartilage of which pharyngeal arch?

- (A) Second pharyngeal arch
- (B) Fourth pharyngeal arch
- (C) Sixth pharyngeal arch
- (D) Third pharyngeal arch

(E) First pharyngeal arch

159. Premature fusion of the sagittal suture produces which skull shape?

- (A) Meroencephaly
- (B) Brachycephaly
- (C) Plagiocephaly
- (D) Scaphocephaly
- (E) Trigonocephaly

160. Premature fusion of the coronal suture on both sides produces which skull shape?

- (A) Meroencephaly
- (B) Brachycephaly
- (C) Trigonocephaly
- (D) Plagiocephaly
- (E) Scaphocephaly

161. Hemivertebra arises from failure of the chondrification centre of which structure?

- (A) Spinous process
- (B) Centrum
- (C) Neural arch
- (D) Costal process
- (E) Transverse process

162. Skeletal muscle mostly develops from which embryonic tissue?

- (A) Splanchnic mesoderm
- (B) Neural crest
- (C) Surface ectoderm
- (D) Paraxial mesoderm
- (E) Lateral mesoderm

163. Smooth muscle mostly develops from which embryonic tissue?

- (A) Notochord
- (B) Surface ectoderm
- (C) Paraxial mesoderm
- (D) Neural tube
- (E) Splanchnic mesoderm

164. Which structure exerts an inductive influence guiding proximodistal limb growth?

- (A) Apical ectodermal ridge
- (B) Neural crest
- (C) Notochord
- (D) Somite core
- (E) Zone of polarizing activity

165. Syndactyly (fused digits) results from failure of which developmental process?

- (A) Vessel outgrowth
- (B) Bone fusion
- (C) Interdigital apoptosis
- (D) Neural migration
- (E) Cartilage synthesis

166. The dermis of the limbs and body wall is derived from which mesoderm?

- (A) Intermediate mesoderm
- (B) Axial mesoderm
- (C) Lateral plate mesoderm
- (D) Paraxial mesoderm
- (E) Splanchnic mesoderm

167. The dermis of the face and neck is derived from which cells?

- (A) Neural crest cells
- (B) Lateral plate mesoderm
- (C) Paraxial mesoderm
- (D) Surface ectoderm
- (E) Splanchnic mesoderm

168. During development the upper limb rotates in which direction?

- (A) Only proximally
- (B) Laterally through ninety degrees
- (C) Fully reversed
- (D) Not at all
- (E) Medially through ninety degrees

169. Congenital torticollis (tilted neck) involves which muscle?

- (A) Trapezius muscle
- (B) Levator scapulae
- (C) Splenius capitis
- (D) Scalene muscle
- (E) Sternocleidomastoid

170. Poland syndrome is characterised by absence of which muscle?

- (A) Sternocleidomastoid
- (B) Abdominal wall muscle
- (C) Serratus anterior
- (D) Pectoralis major
- (E) Pectoralis minor

171. Vernix caseosa on the newborn is produced by which gland?

- (A) Sebaceous gland
- (B) Mammary gland
- (C) Ceruminous gland
- (D) Apocrine gland
- (E) Eccrine gland

Answer Key with Explanations

Keys derived programmatically from concept anchors and verified.

SECTION A -- Histology, Cartilage, Bone, Integument, and Joint Introduction

- 1. B — Sagittal plane**
The sagittal plane divides the body into left and right; coronal into front and back; transverse into upper and lower.
- 2. A — Transverse plane**
The transverse (axial) plane gives the horizontal top-versus-bottom view used by default in computed tomography.
- 3. D — Reposition**
Opposition moves the thumb across the palm; reposition returns it to the anatomical position.
- 4. C — Gomphosis**
A gomphosis is the peg-in-socket fibrous joint of a tooth in its alveolar process.
- 5. E — Syndesmosis**
A syndesmosis unites bones by a sheet of fibrous tissue such as the interosseous membrane.
- 6. B — Synchondrosis**
A synchondrosis is a primary cartilaginous joint of hyaline cartilage, such as the epiphyseal plate.
- 7. E — Symphysis**
A symphysis is a slightly movable secondary cartilaginous joint united by fibrocartilage.
- 8. A — Hinge joint**
A hinge joint is uniaxial, permitting only flexion and extension (for example the elbow).
- 9. C — Saddle joint**
The thumb carpometacarpal joint is a biaxial saddle joint permitting flexion, extension, abduction and adduction.
- 10. D — Fibrocartilage**
A meniscus is a crescent-shaped piece of fibrocartilage that improves fit and absorbs shock.
- 11. A — Strain injury**
A strain injures a muscle or tendon; a sprain injures a ligament.
- 12. B — Bursitis**
Bursitis is inflammation of a bursa; tendinitis is inflammation of a tendon.
- 13. D — Fibroblast**
The fibroblast is the active builder of connective-tissue matrix, with an oval euchromatic nucleus.
- 14. C — Plasma cell**
Plasma cells, derived from B lymphocytes, show clock-face heterochromatin and secrete antibodies.
- 15. E — Mast cell**
Mast cells store histamine and heparin and mediate allergic reactions; they stain metachromatically.
- 16. C — Langerhans cell**
Skin macrophages are Langerhans cells; liver has Kupffer cells; lung alveoli have dust cells.
- 17. B — Neutrophil**
Neutrophils are the first and most powerful phagocytes at bacterial infection sites, dying to form pus.
- 18. A — Eosinophil**
Eosinophils, with bilobed nuclei, increase after parasitic infection and attack parasites.
- 19. D — Glycoproteins**
Ground substance is a gel of proteoglycans (glycosaminoglycans) and multiadhesive glycoproteins.
- 20. E — Basement membrane**
Type IV collagen forms the meshwork of basement membranes and kidney glomeruli.
- 21. D — Type three silver**
Reticular fibres are type III collagen, argyrophilic (silver-loving), and also periodic-acid-Schiff positive.
- 22. C — Fibrillin one**
Marfan syndrome is a fibrillin-1 defect weakening elastic tissue, causing aortic dissection and lens dislocation.
- 23. E — Type three collagen**
Vascular Ehlers-Danlos syndrome is a type III collagen defect causing thin skin, bruising and rupture.
- 24. A — Mucous connective**

Umbilical cord and dental pulp contain mucous connective tissue rich in hyaluronic acid.

25. B — Fibronectin

Fibronectin and laminin link integrins, collagen and proteoglycans to anchor cells to the matrix.

26. E — Mitochondria

Multilocular brown fat has abundant mitochondria for non-shivering thermogenesis in the newborn.

27. C — Two cisternae

A triad is one transverse tubule plus two terminal cisternae of the sarcoplasmic reticulum.

28. D — The isotropic band

The I-band (isotropic band) contains only thin filaments; the A-band holds the thick filaments.

29. A — Postsynaptic receptor

Curare competitively blocks postsynaptic acetylcholine receptors, preventing binding of acetylcholine.

30. B — Presynaptic membrane

Botulinum toxin binds the presynaptic membrane and blocks acetylcholine release.

31. B — Acetylcholine receptor

In myasthenia gravis autoantibodies block postsynaptic acetylcholine receptors, causing fatigable weakness.

32. E — Muscle tension

Golgi tendon organs detect tension and give inhibitory feedback; muscle spindles detect length.

33. D — Muscle spindle

The muscle spindle senses stretch and drives the reflex contraction seen in the knee jerk.

34. C — Ionic coupling

Gap junctions provide ionic coupling for synchronous contraction of cardiac muscle.

35. A — Diad structure

Cardiac muscle has diads (one transverse tubule plus one cisterna) at the Z-line rather than triads.

36. C — Dense bodies

Smooth-muscle dense bodies anchor thin filaments and are the analog of the striated-muscle Z-disc.

37. B — Caveolae

Smooth muscle lacks transverse tubules; caveolae of the sarcolemma trap calcium instead.

38. D — Smooth muscle

Only smooth muscle divides (hyperplasia); skeletal and cardiac muscle enlarge only by hypertrophy.

39. A — Endomysium

Endomysium wraps each fibre; perimysium wraps a fascicle; epimysium wraps the whole muscle.

40. E — Satellite cell

Satellite cells lie under the basal lamina and fuse to repair skeletal muscle after injury.

41. C — Fibrocartilage

Fibrocartilage has type I collagen and no perichondrium, found in intervertebral discs and menisci.

42. B — Orcein stain

Elastic fibres of elastic cartilage stain specifically with orcein.

43. A — Interstitial growth

In interstitial growth chondrocytes divide and secrete matrix within lacunae, lengthening cartilage.

44. E — Appositional growth

Appositional growth adds matrix to the cartilage surface via perichondrial chondroblasts.

45. D — Somatotropin

Growth hormone (somatotropin), thyroxine and testosterone induce cartilage growth; cortisone reduces it.

46. B — Chondroitin sulfate

Territorial matrix is rich in sulfated glycosaminoglycans (chondroitin sulfate), staining basophilic.

47. C — Matrix synthesis

Vitamin C is needed for collagen and matrix synthesis; deficiency causes scurvy.

48. D — Type one collagen

About ninety percent of bone organic matrix is type I collagen.

49. E — Hydroxyapatite

Bone mineral is hydroxyapatite (calcium phosphate).

- 50. A — Monocyte lineage**
Osteoclasts arise from the monocyte-macrophage lineage and resorb mineralised bone.
- 51. C — Stimulate osteoclasts**
Parathyroid hormone increases osteoclast activity and resorption; calcitonin inhibits osteoclasts.
- 52. D — Osteon system**
The osteon (Haversian system) is the unit of compact bone surrounding a central canal.
- 53. A — Volkmann canals**
Volkmann (perforating) canals run transversely, linking Haversian canals and the periosteum.
- 54. E — Irregular fibres**
Woven (primary) bone has irregularly arranged collagen and forms rapidly in the fetus and early repair.
- 55. B — Osteon remnants**
Interstitial lamellae are remnants of old osteons left behind after remodeling.
- 56. C — Hyaline cartilage**
Endochondral ossification replaces a hyaline cartilage model with bone.
- 57. E — Bony callus phase**
Osteoblasts laying down osteoid mark the bony (hard) callus phase, after the fibrocartilaginous callus.
- 58. A — Four to six weeks**
Cancellous bone heals in about four to six weeks; cortical bone takes about eight to twelve weeks.
- 59. D — Resorption exceeds formation**
In osteoporosis osteoclastic resorption exceeds osteoblastic formation, reducing bone mass.
- 60. B — Surface ectoderm**
The epidermis is keratinised stratified squamous epithelium of surface ectodermal origin.
- 61. A — Merkel cell**
Merkel cells in the basal layer act as light-touch receptors.
- 62. B — Antigen presenters**
Langerhans (dendritic) cells of the stratum spinosum are antigen-presenting immune cells.
- 63. C — Stratum lucidum**
The stratum lucidum appears only in thick (glabrous) skin such as the palms and soles.
- 64. E — Stratum granulosum**
The stratum granulosum contains keratohyalin granules and lipid-rich lamellar granules.
- 65. D — Neural crest cells**
Melanocytes arise from neural crest cells and synthesise melanin from tyrosine.
- 66. B — Loose connective tissue**
The papillary dermis is loose connective tissue with capillary loops and Meissner corpuscles.
- 67. E — Dense irregular tissue**
The reticular dermis is dense irregular connective tissue with thick collagen and elastic fibres.
- 68. D — Apocrine gland**
Apocrine glands open into hair follicles in the axilla, areola and anal region.
- 69. A — Thermoregulation**
Eccrine glands are simple coiled tubular glands of thermoregulation, abundant on palms and soles.
- 70. C — Hair follicle**
Sebaceous glands empty sebum into the upper third of the hair follicle.
- 71. B — Smooth sympathetic muscle**
Arrector pili is smooth muscle with cholinergic sympathetic supply, producing goose bumps.
- 72. D — Hair cortex**
The cortex holds melanin and moisture; the medulla is the spongy air-containing core.

SECTION B -- Gross Anatomy (Back, Upper and Lower Limb, Joints, Vertebral Column)

- 73. E — Rhomboid major**
The dorsal scapular nerve supplies the rhomboids and levator scapulae.

- 74. A — Trapezius muscle**
The spinal accessory nerve supplies trapezius and sternocleidomastoid.
- 75. C — Latissimus dorsi**
The triangle of auscultation is bounded by trapezius, rhomboid major and latissimus dorsi.
- 76. D — Dorsal primary rami**
Intrinsic (deep) back muscles are supplied by the dorsal primary rami of spinal nerves.
- 77. B — Superior and inferior obliques**
The suboccipital triangle is bounded by rectus capitis posterior major, obliquus capitis superior and inferior.
- 78. C — Suboccipital nerve**
The suboccipital (first cervical) nerve and the vertebral artery lie within the suboccipital triangle.
- 79. A — Second cervical nerve**
The greater occipital nerve is the dorsal ramus of the second cervical nerve.
- 80. E — Supraspinatus infraspinatus teres minor subscapularis**
The rotator cuff comprises supraspinatus, infraspinatus, teres minor and subscapularis.
- 81. D — Quadrangular space**
The axillary nerve and posterior circumflex humeral vessels traverse the quadrangular space.
- 82. C — Axillary nerve**
The axillary nerve supplies the deltoid and teres minor.
- 83. E — Suprascapular nerve**
The suprascapular nerve supplies supraspinatus and infraspinatus.
- 84. A — One to three**
Scapular to glenohumeral movement is roughly one to three during abduction.
- 85. B — Pectoralis major**
Pectoralis major forms the anterior axillary fold; latissimus dorsi and teres major form the posterior fold.
- 86. A — Pectoralis major**
The deltopectoral triangle lies between the clavicle, deltoid and superior border of pectoralis major.
- 87. B — Axillary vein**
The cephalic vein passes through the deltopectoral triangle to join the axillary vein.
- 88. E — Medial pectoral nerve**
The medial pectoral nerve supplies pectoralis minor and the lower part of pectoralis major.
- 89. C — Long thoracic nerve**
The long thoracic nerve supplies serratus anterior; injury causes a winged scapula.
- 90. D — Musculocutaneous nerve**
The musculocutaneous nerve supplies the arm flexors: biceps, brachialis and coracobrachialis.
- 91. E — Radial groove**
The radial nerve runs in the radial (spiral) groove with the profunda brachii artery.
- 92. A — Brachial artery**
The brachial artery lies in the cubital fossa deep to the bicipital aponeurosis, medial to the biceps tendon.
- 93. B — Brachioradialis muscle**
Brachioradialis forms the lateral border; pronator teres forms the medial border of the cubital fossa.
- 94. D — Radial tuberosity**
The biceps tendon inserts on the radial tuberosity; brachialis inserts on the coronoid and ulnar tuberosity.
- 95. C — Medial epicondyle**
Pronator teres, flexor carpi radialis, palmaris longus and flexor carpi ulnaris share the common flexor origin.
- 96. C — Lateral epicondyle**
The common extensor origin is the lateral epicondyle; overuse causes tennis elbow.
- 97. E — Median nerve**
Pronator teres is a median-nerve muscle inserting on the mid-shaft of the radius.
- 98. B — Wrist drop**
Radial nerve compression in the arm causes wrist drop from loss of wrist and finger extension.
- 99. A — Radius bone**
A Colles fracture is a distal radius fracture with dorsal displacement of the fragment.

100. D — Trapezium bone

The snuffbox floor is the scaphoid and trapezium; its content is the second part of the radial artery.

101. A — Second part of radial artery

The second part of the radial artery crosses the floor of the snuffbox.

102. C — Superficial palmar arch

The central palmar compartment contains flexor tendons, lumbricals and the superficial palmar arch.

103. E — Palmaris longus tendon

The palmar aponeurosis continues from the palmaris longus tendon over the flexor retinaculum.

104. B — Six tunnels

The extensor retinaculum forms six osseofibrous tunnels transmitting nine extensor muscle tendons.

105. D — Saddle joint

The sternoclavicular joint is a saddle-type synovial joint.

106. D — Condylod joint

The metacarpophalangeal joints are biaxial condylod joints allowing flexion, extension, abduction and adduction.

107. B — Plane joint

The acromioclavicular joint is a plane (gliding) synovial joint.

108. A — Glenoid labrum

The glenoid labrum deepens the shallow glenoid cavity of the ball-and-socket shoulder.

109. C — Annular ligament

The annular ligament encircles the radial head at the proximal radioulnar joint.

110. E — Gluteus maximus

Gluteus maximus extends and laterally rotates the thigh; it is supplied by the inferior gluteal nerve.

111. A — Superior gluteal nerve

The superior gluteal nerve supplies gluteus medius, minimus and tensor fasciae latae.

112. D — Superior gluteal nerve

Superior gluteal nerve palsy weakens the hip abductors, giving a Trendelenburg gait.

113. E — Ischial tuberosity

The hamstrings (except the short head of biceps femoris) arise from the ischial tuberosity.

114. B — Tibial nerve

The tibial division of the sciatic nerve supplies the hamstrings; the short head of biceps femoris is common fibular.

115. C — Greater trochanter

A line from the posterior superior iliac spine to the greater trochanter marks the safe upper outer zone.

116. D — Tibial nerve

The tibial nerve is the most superficial major structure in the popliteal fossa.

117. A — Biceps femoris

The superolateral boundary of the popliteal fossa is biceps femoris.

118. B — Popliteal vein

The small saphenous vein drains into the popliteal vein.

119. E — Femoral nerve

The femoral nerve supplies the anterior (extensor) thigh compartment.

120. C — Obturator nerve

The obturator nerve supplies the medial adductor compartment of the thigh.

121. C — Adductor longus

The femoral triangle is bounded by the inguinal ligament, sartorius and adductor longus.

122. B — Saphenous nerve

The adductor canal transmits the femoral artery and vein, the saphenous nerve and the nerve to vastus medialis.

123. D — Rectus femoris

Rectus femoris crosses two joints, flexing the hip and extending the knee.

124. E — Loose tissue and lymphatics

The femoral canal holds loose connective tissue, fat and lymphatics, allowing venous distension.

125. A — Deep fibular nerve

The deep fibular nerve supplies the anterior compartment dorsiflexors; its loss causes foot drop.

126. A — Superficial fibular nerve

The superficial fibular nerve supplies fibularis longus and brevis (eversion).

127. E — Tibial nerve

The tibial nerve and posterior tibial artery supply the posterior compartment plantarflexors.

128. D — Tibialis posterior

Tibialis posterior inverts and plantarflexes the foot and supports the arches.

129. B — Femoral vein

The great saphenous vein passes through the saphenous opening to join the femoral vein.

130. C — Spring ligament

The plantar calcaneonavicular (spring) ligament supports the head of the talus and the medial arch.

131. C — Inversion and eversion

The transverse tarsal (and subtalar) joints permit inversion and eversion of the foot.

132. D — Ball and socket joint

The hip is a multiaxial ball-and-socket joint.

133. A — Iliofemoral ligament

The Y-shaped iliofemoral ligament is the strongest and limits hip hyperextension.

134. B — Ischiofemoral ligament

The posterior ischiofemoral ligament is torn in posterior hip dislocation.

135. E — Anterior sliding

The anterior cruciate ligament prevents anterior sliding of the tibia on the femur.

136. E — Posterior sliding

The posterior cruciate ligament prevents posterior sliding of the tibia; it is torn in dashboard injuries.

137. D — Fibrocartilage

The menisci are crescent-shaped fibrocartilage that deepen the tibial surface and absorb shock.

138. C — Tibial collateral ligament

The medial meniscus adheres to the tibial (medial) collateral ligament, so both are commonly torn together.

139. A — Popliteus muscle

Popliteus laterally rotates the femur on the tibia to unlock the fully extended knee.

140. B — Hinge joint

The talocrural joint is a hinge permitting dorsiflexion and plantarflexion.

141. C — Anterior talofibular ligament

The anterior talofibular ligament is closely related to the talar neck and is commonly injured.

142. D — Semimembranosus bursa

The semimembranosus bursa usually does not communicate with the knee cavity, unlike the suprapatellar, popliteus and gastrocnemius bursae.

143. B — Hyperextension

The anterior longitudinal ligament prevents hyperextension and is torn in whiplash injuries.

144. E — Posterior longitudinal ligament

The posterior longitudinal ligament resists hyperflexion and helps prevent posterior disc herniation.

145. A — Ligamentum flavum

The elastic ligamentum flavum connects adjacent laminae and limits abrupt flexion; its hypertrophy causes spinal stenosis.

146. C — Plane joint

Facet joints are plane synovial joints permitting gliding between articular processes.

147. A — Pivot joint

The median atlantoaxial joint is a pivot joint where the dens rotates within a ring.

148. E — Transverse ligament of atlas

The transverse ligament of the atlas holds the dens; its rupture causes atlantoaxial subluxation.

149. D — Posterolaterally

The annulus fibrosus is thin posterolaterally, so herniation usually extends in that direction.

150. B — The fifth lumbar or first sacral

About ninety-five percent of lumbar herniations occur at L4-L5 or L5-S1, compressing the L5 or S1 root.

151. B — Scoliosis

Scoliosis is an abnormal lateral curvature; kyphosis and lordosis are exaggerated sagittal curves.

152. D — Thoracic and sacral kyphoses

The thoracic and sacral kyphoses are primary curvatures; the cervical and lumbar lordoses are secondary.

SECTION C -- Embryology and Development

153. E — Notochord

The nucleus pulposus is derived from the notochord.

154. C — Sclerotome

The sclerotome (ventromedial somite) forms the vertebrae and ribs.

155. A — Sphenoid bone

The chondrocranium includes the occipital, sphenoid, ethmoid and petromastoid temporal bones.

156. C — Frontal bone

The membranous neurocranium includes the frontal, parietal and interparietal part of the occipital bone.

157. E — First pharyngeal arch

Meckel's cartilage of the first pharyngeal arch forms the malleus and incus.

158. A — Second pharyngeal arch

Reichert's cartilage of the second pharyngeal arch forms the stapes and styloid process.

159. D — Scaphocephaly

Sagittal synostosis produces a long, narrow, wedge-shaped skull (scaphocephaly).

160. B — Brachycephaly

Bilateral coronal synostosis gives a high, tower-like (short broad) skull, brachycephaly.

161. B — Centrum

Hemivertebra results from failure of a chondrification centre of one half of the centrum.

162. D — Paraxial mesoderm

Skeletal muscle develops mainly from the myotomes of the paraxial mesoderm (somites).

163. E — Splanchnic mesoderm

Smooth muscle develops mostly from splanchnic mesoderm.

164. A — Apical ectodermal ridge

The apical ectodermal ridge drives proximodistal outgrowth; the zone of polarizing activity patterns the anteroposterior axis.

165. C — Interdigital apoptosis

Failure of programmed cell death in the interdigital webs leaves fused digits (syndactyly).

166. C — Lateral plate mesoderm

Limb and body-wall dermis derives from lateral plate mesoderm; back dermis from paraxial mesoderm.

167. A — Neural crest cells

The dermis of the face and neck develops from neural crest cells.

168. B — Laterally through ninety degrees

The upper limb rotates laterally about ninety degrees; the lower limb rotates medially.

169. E — Sternocleidomastoid

Congenital torticollis results from tearing/fibrosis of the sternocleidomastoid during birth.

170. D — Pectoralis major

Poland syndrome features absence of the sternocostal part of pectoralis major, often with syndactyly.

171. A — Sebaceous gland

Sebaceous glands produce the sebum-rich vernix caseosa protecting fetal skin.